

SAKUNA HARINDA JAYASUNDARA

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Auckland, New Zealand

SUMMARY

Engineer and Researcher with 5+ years of industry and research experience building scalable, production-grade systems across backend development, machine learning, and automation. Brings 8+ years of experience in Python, Java, and C/C++, C#, TypeScript, along with extensive work across agentic AI, agentic security, modern ML frameworks, CI/CD pipelines, and test automation. PhD from the University of Auckland, with research spanning deep learning, natural language processing, cybersecurity, and HCI. Focused on applying data-driven and research-informed approaches to solve real-world engineering problems.

EXPERIENCE

• Professional Teaching Fellow

University of Auckland 

March 2026 - Present
Auckland, New Zealand

- Prepare and deliver lectures for undergraduate and postgraduate courses: Data structures and Algorithms, Theoretical computer science, Cryptographic management, and Programming for industry (Java).
- Supervise examinations and assess assignments and projects, ensuring fair and consistent evaluation.

• Software Engineer

H2O.ai 

June 2022 - October 2022
Remote

- Built and maintained scalable [Wave](#) applications to support internal and external ML workflows, enhancing reliability and developer productivity through automation.
- Designed and implemented robust end-to-end test automation pipelines using Python, Jenkins, and Groovy, reducing manual testing effort and increasing release velocity.
- Led the architecture and development of a best-in-class test automation framework, including guiding implementation, defining coding standards, and shaping the automation roadmap for cross-team use.
- Collaborated with engineering teams to integrate automation into CI/CD workflows, improving code quality and shortening time-to-deployment for key features.

• Software Engineer

Axiata Digital Labs 

June 2021 - June 2022
Colombo, Sri Lanka

- Designed, developed, and maintained backend RESTful and SOAP APIs using Java and Spring Boot to support customer-facing services for clients associated with Celcom Malaysia, ensuring high reliability and seamless integration with existing telecommunications systems.
- Built an anomaly detection system leveraging open telemetry data and machine learning models to identify performance and reliability issues in production services, enabling proactive issue detection and improved system observability.
- Developed an NLP-based repository classifier to automatically identify public GitHub projects with potential relevance to internal initiatives, streamlining competitive analysis and reducing manual review workload.

• Intern Electronics Engineer

Paraqum Technologies 

June 2019 - December 2019
Colombo, Sri Lanka

- Developed a GTP (GPRS Tunnelling Protocol) packet analysis tool along with a test environment, enabling detailed inspection and debugging of mobile core network traffic for protocol validation and performance analysis.
- Designed and implemented a load-balancing software module to monitor network interfaces and manage data traffic efficiently, improving throughput handling in simulated network scenarios.
- Enhanced performance and extended functionality of the company's AD client software, optimising existing C++ modules for responsiveness and reliability under real-world conditions.

EDUCATION

- **Ph.D. in Computer Science**

The University of Auckland

October 2022 - January 2026

Auckland, New Zealand

- Introduced a human-centric approach to automated access control policy generation, addressing fundamental limitations of both manual and fully automated methods by reducing human error and AI-induced inaccuracies.
- Developed language model-driven techniques to translate natural-language access requirements into enforceable access control policies, enabling accurate handling of complex, ambiguous, real-world specifications.
- Designed ML-based mechanisms that support automatic error detection and recovery, enabling safe deployment of AI-generated security policies in high-risk environments.
- Conducted empirical user studies with professional policy implementers, demonstrating improved accuracy, efficiency, trust, and adoption potential compared to existing approaches.
- Resulted in three publications (one in a flagship cybersecurity conference and two in Q1 computer science journals) with two under review in flagship computer science and HCI conferences (See [Google Scholar](#) for the full list of publications.).

- **B.Sc. (Hons.) in Electronics and Telecommunication Engineering**

The University of Moratuwa

November 2016 - July 2021

Colombo, Sri Lanka

- Attained a GPA of 3.92/4.2, obtaining a first class.
- Dean's List Honouree, recognised in all four academic years of the B.Sc. program.
- Developed a deep learning-based maritime surveillance software for the Sri Lanka Navy to detect suspicious activities in the sea using PyTorch and PyQt. The project won the 2nd runner-up in the National ICT Awards for the best university project.

TECHNICAL SKILLS

- **Knowledge Areas:** Software Engineering, DevOps, Machine Learning, Computer Vision, Natural Language Processing & Signal Processing
- **Programming Languages:** Python, Java, C/C++, C#, GO, & Dart
- **Web Technologies:** HTML, CSS, TypeScript, & JavaScript
- **Database Systems:** SQL, MongoDB, & Neo4J
- **Software & Machine Learning Frameworks:** Langchain, Langgraph, CrewAI, Autogen, Google ADK, Springboot, FastAPI, Pytorch, Keras, Tensorflow, & Scikit-learn
- **Cloud Technologies:** AWS & Azure
- **DevOps & Version Control:** Docker, Kubernetes, Jenkins, Git, GitHub, & GitHub actions
- **API Protocols:** REST, JSONRPC, & SOAP
- **Software Development Methodologies:** Agile

HONOURS AND AWARDS

- **ESET Chillisoft Scholarship in Cybersecurity** *November 2022*
ESET
- **MBIE Artificial Intelligence for Human-Centric Security Project Scholarship** *October 2022*
Ministry of Business, Innovation and Employment (MBIE), New Zealand
- **2nd runner-up in National ICT Awards - Best university project** *March 2021*
NBQSA, Sri Lanka

KNOWLEDGE SHARING AND TEACHING

- **Graduate Teaching Assistant** *December 2022 - May 2025*
The University of Auckland
 - Delivered tutorials and laboratory sessions on Java programming and web technologies for postgraduate courses.
 - Supported students through hands-on guidance in coursework, labs, and project-based assignments.
- **Tutor/Workshop Instructor** *March 2021 - March 2023*
The University of Moratuwa
 - Delivered lectures and technical workshops on the Robot Operating System (ROS) as part of the IEEE RAS Sri Lanka Chapter seminar series.
 - Designed and facilitated hands-on exercises to reinforce core robotics concepts and practical ROS workflows.

REFERENCES

References will be available upon request